Interview Prep for Backend Developer

this is my resume:

Sayan Ghosh

AI-proof Problem Solver | B.Tech in Computer Science Engineering | KIIT University, Bhubaneswar

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Work Experience

AICTE-EduSkills & Google | AI-ML Virtual Intern

Completed a hands-on program focusing on Object Detection, Image Classification, and Product Image Search using TensorFlow and Google AI tools.

Earned 7 Google Developer Badges for expertise in Computer Vision and Deep Learning.

Gained practical experience with Google Cloud AI, Model Training, and Deployment, enhancing model accuracy and efficiency in real-world applications.

Infosys | AI Intern

Developed MediTrain, an AI-powered medical chatbot using Groq AI, improving accuracy by 30% and reducing misclassification by 20%.

Designed a context-aware AI system, increasing user engagement by 25%.

Built a secure, API-integrated chatbot interface for real-time medical information retrieval.

Led Agile sprints, driving weekly model iterations and API optimizations, reducing deployment cycles by 40%.

Proposed multilingual support and enhanced conversation memory to lay the groundwork for global scalability.

AICTE-EduSkills & AWS Academy | Cloud Virtual Intern

Gained expertise in AWS cloud technologies, including EC2, S3, RDS, VPC, and Auto Scaling.

Optimized cloud resource allocation, reducing costs by 20% in dynamic workloads.

Implemented IAM policies, encryption, and VPC isolation, strengthening security and reducing vulnerabilities by 35%.

Designed high-availability architectures, ensuring 99.9% uptime in theoretical scenarios.

Technical Skills

Programming Languages: C, Python (AI/ML), C++ (DSA)

Developer Tools: CI/CD Pipeline, Git/GitHub, Jenkins, Docker, Postman, Linux, PostgreSQL, MySQL

Expertise: Machine Learning, Deep Learning, Back-end Development, Logic Building

Projects

CarTagDetect | GitHub | YouTube Demo

Built a Vehicle and License Plate Detection System using YOLOv8, EasyOCR, and SORT, achieving 92% vehicle tracking and 85% license plate accuracy.

Integrated data interpolation, annotated overlays, and automated CSV export with timestamps for traffic monitoring and security applications.

Tech Stack: Python, YOLOv8, EasyOCR, SORT, OpenCV, SciPy, NumPy

FoxxData - Synthetic Data Generation | GitHub

Developed an LLM-driven data augmentation pipeline using meta-prompt optimization and Few-Shot Prompting for fraud detection.

Engineered dynamic token interpolation and adaptive schema enforcement via Pydantic serialization to enhance data integrity.

Integrated multi-stage parsing heuristics to ensure statistical coherence in synthetic data.

Tech Stack: Python, LangChain, Google Gemini API, Pydantic, Prompt Engineering

Education

Kalinga Institute of Industrial Technology, Bhubaneswar

B.Tech in Computer Science Engineering | CGPA: 8.17/10

Relevant Coursework: DBMS, Operating Systems (OS), OOPS, Computer Networks (CN), Software Engineering (SE), Computer Architecture (COA)

Certifications

Applied AI with DeepLearning by IBM

ML Operations Specialization by Duke University

Additional Achievements

Published an Edge add-on on the Microsoft Store with 1,000+ users, enabling web page sharing across devices via QR codes with zero technical configuration.

This is the Job description for the position:

Job Description

Backend Developer Intern

WhatYou'll Do

 Developscalable backend APIs and microservices using Java Spring Boot.

 Work with SQL (MySQL, PostgreSQL) & NoSQL (MongoDB, Redis) databases for efficient

data storage and retrieval.

 Optimize backend performance to handle high-concurrent transactions.

 Implementauthentication, authorization, and security best practices.

 Collaborate with frontend developers, DevOps, and product teams to deliver seamless

applications.

 Participate in code reviews, debugging, and performance tuning.

 Gainexposureto cloud-based deployment and CI/CD pipelines.

 Understand EV charging network integrations and how backend systems power real-time

EV charging in India and Dubai.

 Get immersed in the UAE market to understand how backend solutions support international operations.

WhoShouldApply?

 Proficiency in Java & Spring Boot Framework with a deep understanding of RESTful API

development.

 Strong knowledge of SQL databases (MySQL, PostgreSQL) and NoSQL (MongoDB, Redis,

DynamoDB).

 Understanding of Microservices architecture and API security best practices.

 Basicfamiliarity with cloud services (AWS, GCP, or Azure) is a plus.

 Self-driven, eager to learn, and passionate about building scalable backend architectures.

What's in it for You?

 Directmentorship from senior backend engineers & CTO, working on real-world projects.

 Exposure to the UAE market, learning how backend technologies support international

business operations.

 Workinahigh-growth, tech-driven startup, where innovation and problem-solving are

encouraged.

Now, I have to prepare for an interview at Zynetics for a backend developer intern. Help me prepare for it by giving me a lot of questions along with their answers that might come in the interview.

Great! Based on your resume and the job description for the Backend Developer Intern role at Zynetics, here's a curated list of potential interview questions and answers to help you prepare. The questions cover core backend concepts, Java Spring Boot, databases, cloud, security, and your past experiences.

🔧 Technical Questions (Java, Spring Boot, API)

1. What is Spring Boot and how is it different from Spring Framework?

Answer:
Spring Boot is an extension of the Spring framework that simplifies the setup and development of new Spring applications. Unlike Spring, which requires a lot of boilerplate code and configuration, Spring

Boot uses convention over configuration, auto-configures dependencies, and comes with embedded servers like Tomcat.

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2. How do you create a REST API in Spring Boot?

Answer:
You annotate a class with `@RestController` and define routes using `@GetMapping`, `@PostMapping`, etc. For example:

```java
`@RestController
@RequestMapping("/api")
public class SampleController {
    @GetMapping("/hello")
    public String sayHello() {
        return "Hello World";
    }
}
`
```

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3. What is Dependency Injection in Spring Boot?

Answer:
Dependency Injection is a design pattern that allows a class to receive its dependencies from external sources rather than creating them itself. In Spring Boot, this is handled using annotations like `@Autowired`.

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4. What is the difference between `@Component`, `@Service`, and `@Repository`?

Answer:
All three are Spring stereotypes for different layers:

- `@Component`: Generic component.
- `@Service`: Business logic layer.
- `@Repository`: DAO (Data Access Object) layer with additional persistence exception translation.

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5. What are some best practices for securing REST APIs?

Answer:

- Use HTTPS
- Validate inputs
- Use tokens (JWT) for authentication
- Implement role-based access control (RBAC)
- Use API gateways and rate limiting

🗃️ Database & SQL/NoSQL

6. What is the difference between SQL and NoSQL databases?

Answer:

SQL (Relational)	NoSQL (Non-relational)
Structured schema	Dynamic or flexible schema
Use SQL for queries	Use JSON-like queries
Good for ACID compliance	Good for scalability & speed
Examples: MySQL, PostgreSQL	MongoDB, Redis

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7. What are joins in SQL?

Answer:
Joins are used to retrieve data from two or more tables:

- `` `INNER JOIN` ``: Common records
- `` `LEFT JOIN` ``: All from left + matches from right
- `` `RIGHT JOIN` ``: All from right + matches from left
- `` `FULL JOIN` ``: All records when there is a match in either table

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8. What are indexes in a database?

Answer:
Indexes improve the speed of data retrieval operations. They work like an index in a book — allowing the DB engine to quickly locate data without scanning the entire table.

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9. How do you prevent SQL injection?

Answer:

- Use Prepared Statements or ORM frameworks like JPA
- Validate user inputs
- Avoid dynamic SQL queries

☁️ Cloud & DevOps

10. What is CI/CD and how have you used it?

Answer:
CI/CD stands for Continuous Integration and Continuous Deployment. It automates the process of testing, building, and deploying code. I've used Jenkins, GitHub Actions, and Docker to automate deployments during my projects and internships.

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11. Explain the components of AWS you've used.

Answer:

- **EC2:** Virtual machines
- **S3:** Object storage
- **RDS:** Relational database service
- **IAM:** Access management
- **VPC:** Virtual network
- I used these to build secure, scalable apps and reduce costs by optimizing compute/storage usage.

🧠 Conceptual / Backend Architecture

12. What are Microservices?

Answer:
Microservices are an architectural style where applications are composed of small, independent services that communicate via APIs. Each service is responsible for a specific business function and can be deployed independently.

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13. What is load balancing?

Answer:
Load balancing distributes network or application traffic across multiple servers to ensure no single server is overwhelmed, improving responsiveness and availability.

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14. What are some common HTTP methods used in REST APIs?

Answer:

- `GET`: Retrieve data
- `POST`: Create data
- `PUT`: Update data
- `DELETE`: Delete data
- `PATCH`: Partially update data

🧑‍💼 Behavioral & Project-Based Questions

15. Tell me about your project FoxxData.

Answer:
FoxxData is an LLM-powered synthetic data generation tool. I used meta-prompt optimization and token

interpolation to generate fraud detection datasets. Pydantic enforced schema integrity, and LangChain powered the pipeline. It's designed to enhance AI model performance where real data is scarce.

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16. Describe a challenge you faced during your Infosys internship.

Answer:
One challenge was reducing misclassification in the MediTrain chatbot. I introduced a context-aware module and improved the intent recognition pipeline, which reduced misclassification by 20% and improved engagement.

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17. Why are you interested in Zynetics and backend development?

Answer:
I love backend engineering because it combines logic, performance tuning, and system design. Zynetics excites me due to its real-world impact in EV infrastructure and exposure to the UAE market, which aligns with my goal to build globally scalable systems.

🧩 System Design (Basic)

18. How would you design a URL shortener (like bit.ly)?

Answer:

1. Accept long URL → generate a unique short key.
2. Store mapping in DB: short key → long URL.
3. Use base62 encoding for key generation.
4. Add expiration logic if needed.
5. Use caching (Redis) for faster retrieval.

Would you like:

- **Mock interview style Q&A?**
- **More system design problems?**
- **Behavioral questions tailored to Zynetics culture?**

Let me know how you want to drill deeper!